



1 Vector and phasor representation

- a) What is the magnitude $|\mathbf{E}_1|$ of the field vector $E_1 = -2\mathbf{x}_0 + 4\mathbf{y}_0 + \mathbf{z}_0$?
- b) What is the sum of the two field vectors $\mathbf{E}_3 = \mathbf{E}_1 + \mathbf{E}_2$ when $\mathbf{E}_2 = \mathbf{y}_0 + \mathbf{z}_0$?
And what is the product of the two, $W = \mathbf{E}_1 \cdot \mathbf{E}_2$?
- c) What is the cross product $\mathbf{P} = \mathbf{E}_1 \times \mathbf{H}_1$ when $\mathbf{H}_1 = \mathbf{x}_0 - 2\mathbf{y}_0$?
- d) What is a phasor notation? Write up the phasor notation that can represent the time-harmonic function.

2 Maxwells equations

Classical electrodynamics is described by Maxwell's equations

$$\nabla \cdot \mathbf{E} = \frac{\rho_{\text{tot}}}{\epsilon_0}, \quad (1a) \quad \nabla \cdot \mathbf{D} = \rho, \quad (2a)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (1b) \quad \nabla \cdot \mathbf{B} = 0, \quad (2b)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (1c) \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (2c)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_{\text{tot}} + \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (1d) \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (2d)$$

Refresh your excellent knowledge of Maxwell's equations and state what the terms in Eqs. (1) and (2) are. Name the equations in (1) and give a *short* explanation of what they describe.

3 EM waves in a source-free medium

In this problem we consider a linear, isotropic, and homogeneous medium without sources.

- a) What is a plane wave? What is a TEM wave?
- b) If a waveform is described by $f(t, z) = A \cos(kz - \omega t)$, in what direction does the wave travel?
Set $k = \omega = 1$ and draw the waveform for a few values of t on the range $z \in [-2\pi, 2\pi]$.
Do the same for the waveform $f(t, z) = A \cos(kz + \omega t)$. In what direction does this wave travel?
- c) Write up Maxwell's equations (1) in a vacuum and charge-free space.
- d) Derive the wave equations from these by taking the curl of the equations and using the vector identities

$$\nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - \nabla^2 \mathbf{V} \quad (3)$$

and

$$\nabla^2 \mathbf{V} = \nabla \cdot (\nabla \mathbf{V}) \quad (4)$$

for vector $\mathbf{V}$. What is this special condition of the wave equation with zero source currents called?

- e) Write the Helmholtz equation for the electric field component in the x -direction, E_x . Assume that E_x is only dependent on z . Find the general solution to the equation.
- f) One solution to the Helmholtz equation is given as

$$E_x(z) = E_0^+ \exp(-jkz). \quad (5)$$

What is the physical electric field (in the time domain)?

4 EM Waves in lossy media

- a) What is the parameter $\tan \delta$ which describes the loss in dielectrics? For what values of $\tan \delta$ is the material in question a good dielectric, and when is it a poor dielectric?
- b) What is the main factor contributing to thermal loss in metals?
- c) Draw the propagation of a sine wave in a lossy medium experiencing an attenuation factor α .